HVPS V3 Controller

Preliminary Design Review

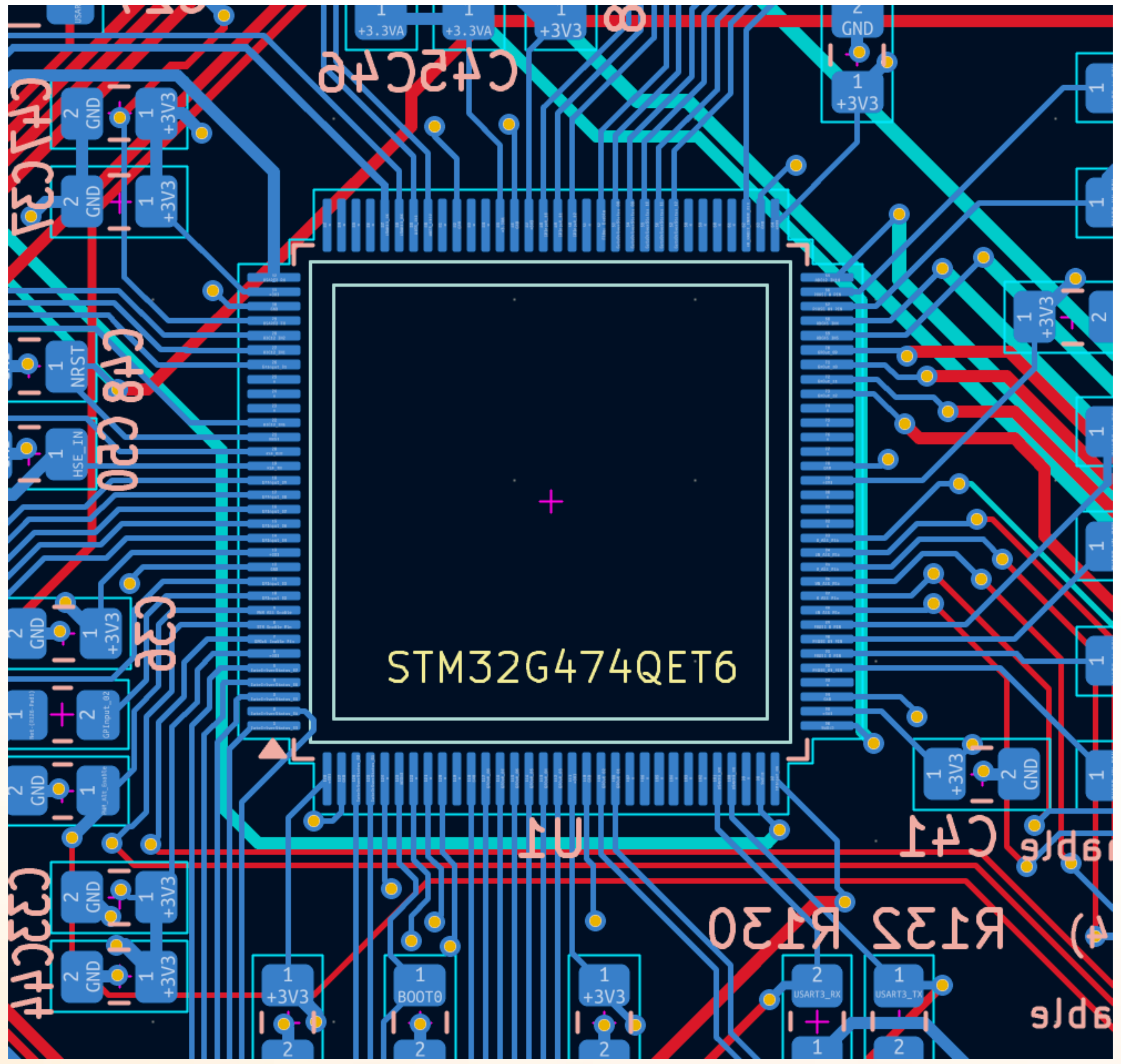
Everett Penne, Feb 26 ‘26

Requirements

- Maintain & improve upon V2 controller functionality
  - 3-phase PWM near 20kHz
    - More adjustable dead time
    - Better output on/off control
  - Improved EMI compatibility
    - MCU directly on-board
    - Increased digital & analog I/O protection

MCU Selection

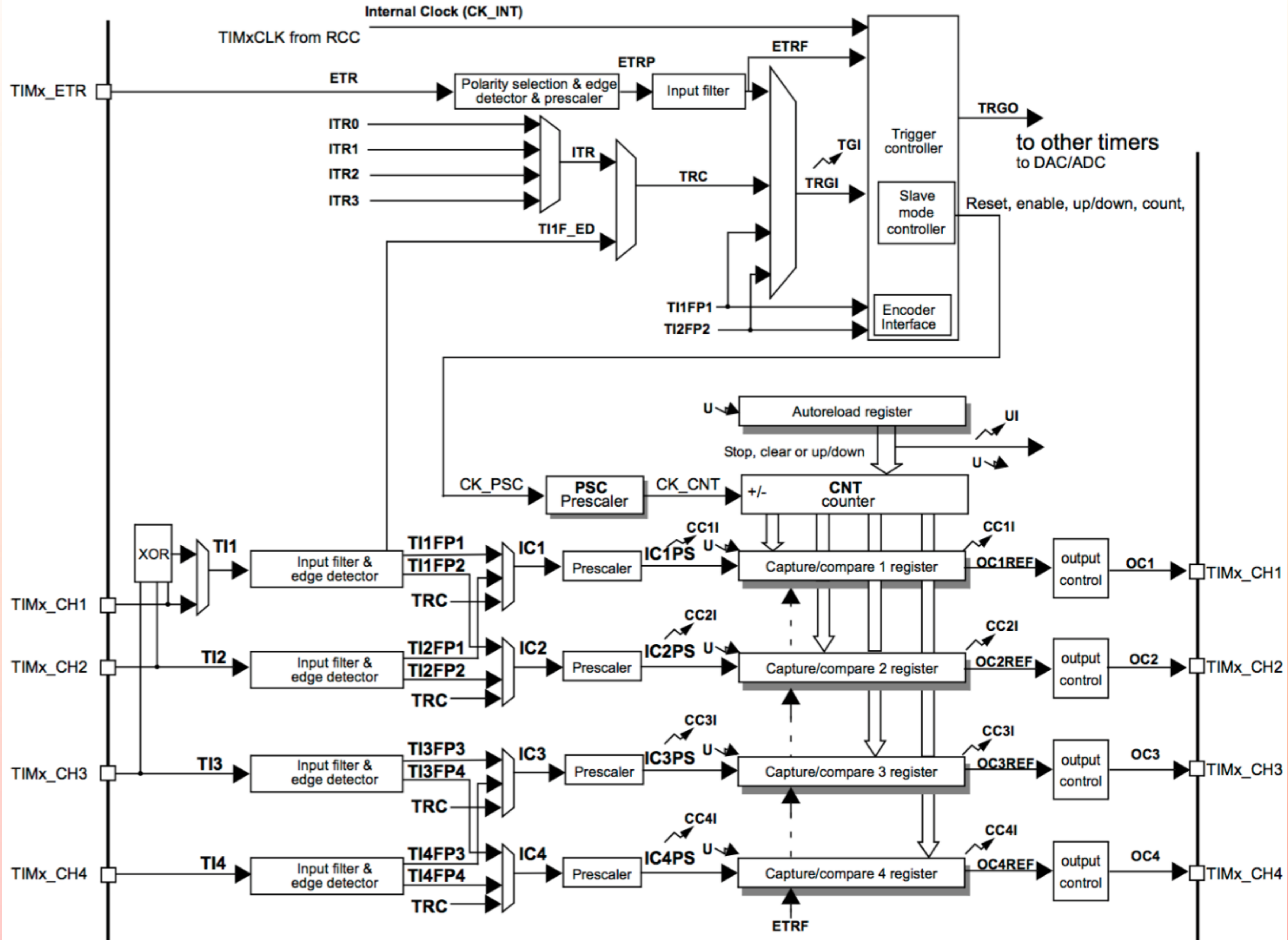
STM32G474QET6



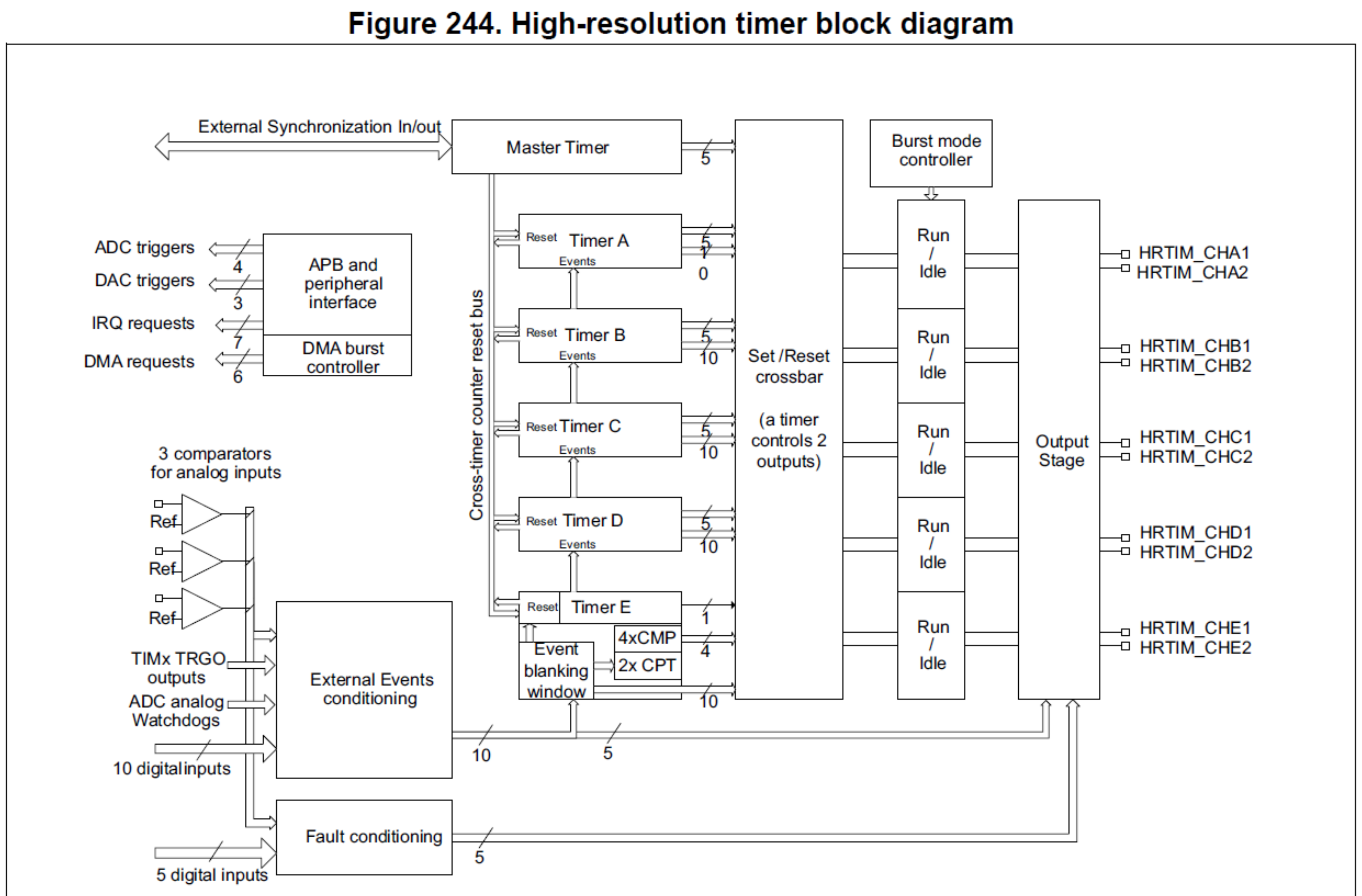
- 170 MHz clock (V2 STM32L4 has 80MHz clock)
- HRTIM functionality
  - 12 HR PWM outputs configurable for up to 6 complementary PWM outputs (eg: A, AN, B, BN, C, CN)
- Multiple 12-bit ADCs (we need 8 channels)
- 107 available GPIO pins (including ADC & Timer pins)
- 3 full USART serial communication channels

V2 PWM Imperfections

HRTIM (High-Resolution Timer)



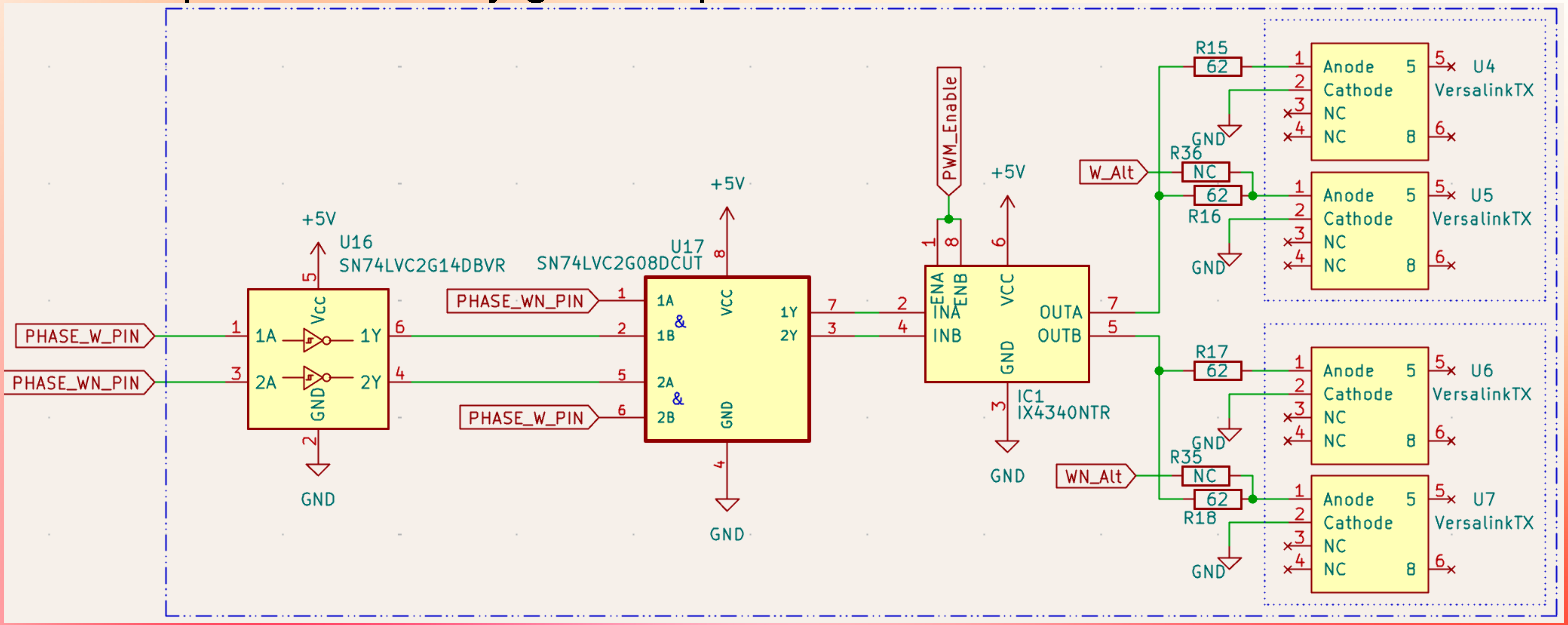
STM32L4 Timer Block diagram (V2 Controller)



STM32G4 HR Timer Block diagram (Proposed V3 Controller)

Better Output On/Off Control

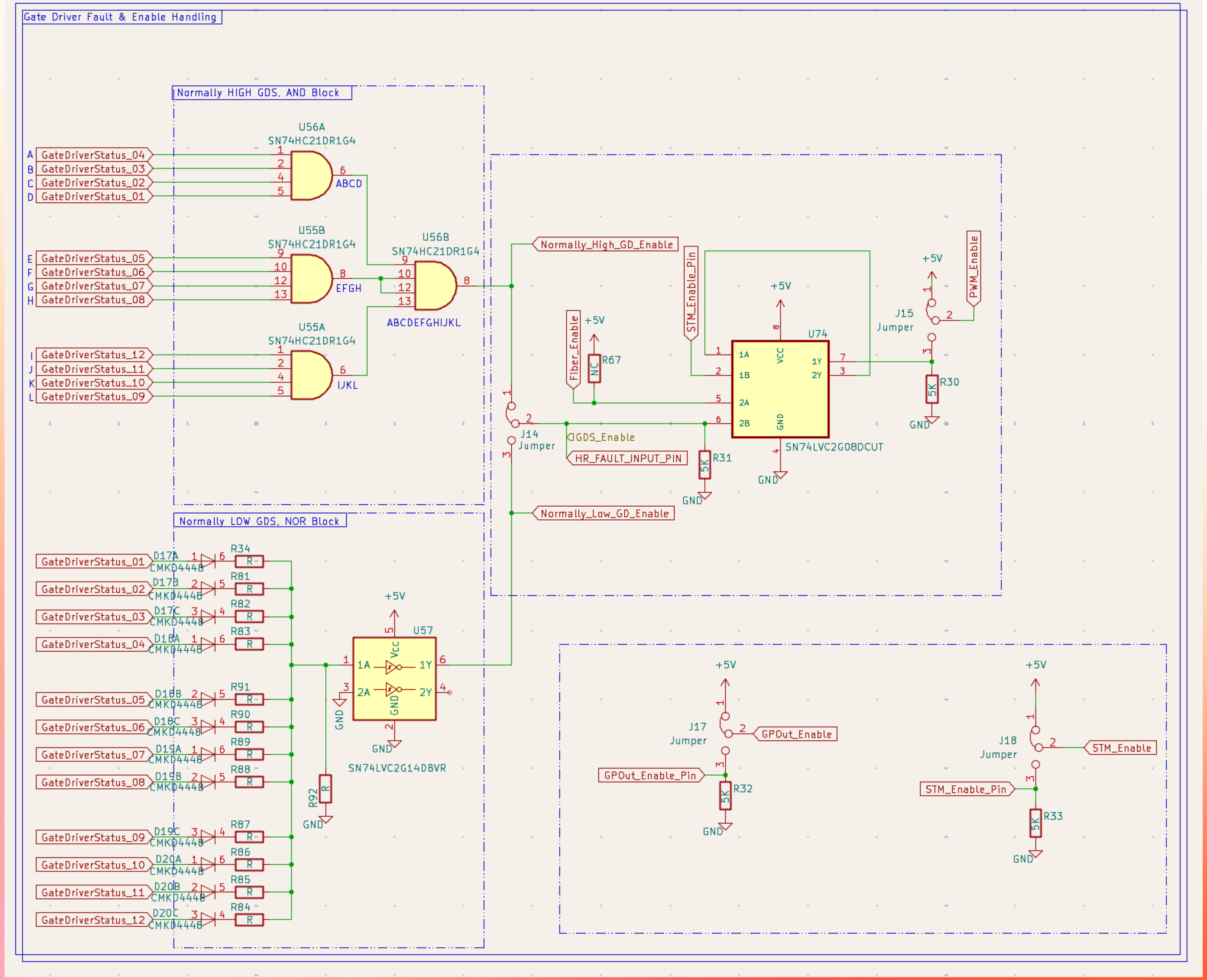
- Connect PWM Fiber Output Gate Driver enable to pin + pull-down resistor on MCU
  - Adds option to directly gate output with software



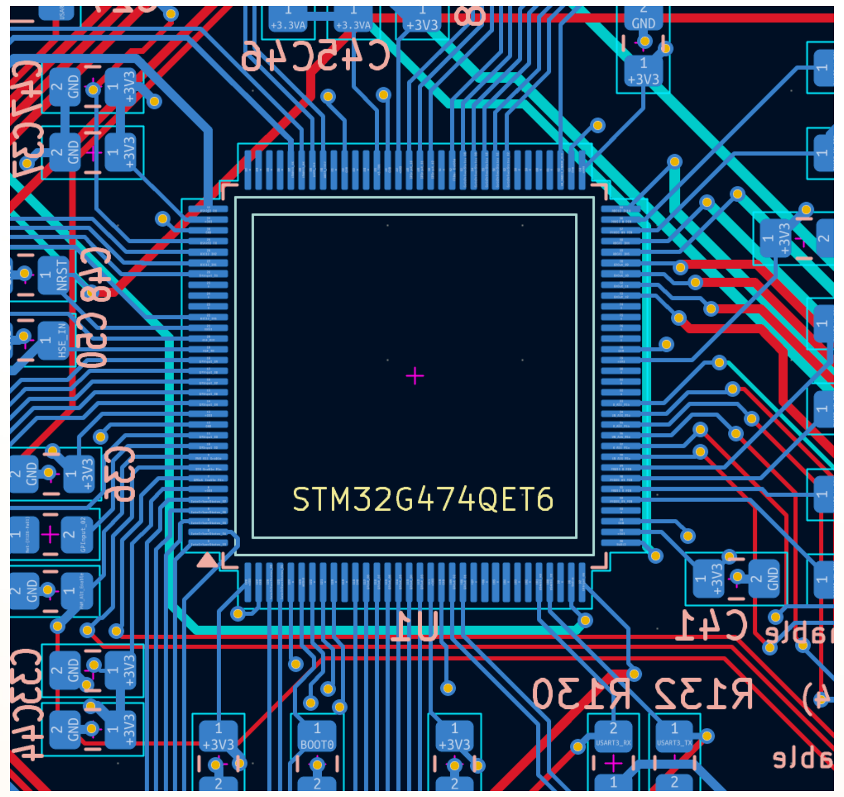
GDStatus Fault Handling

Many options now available

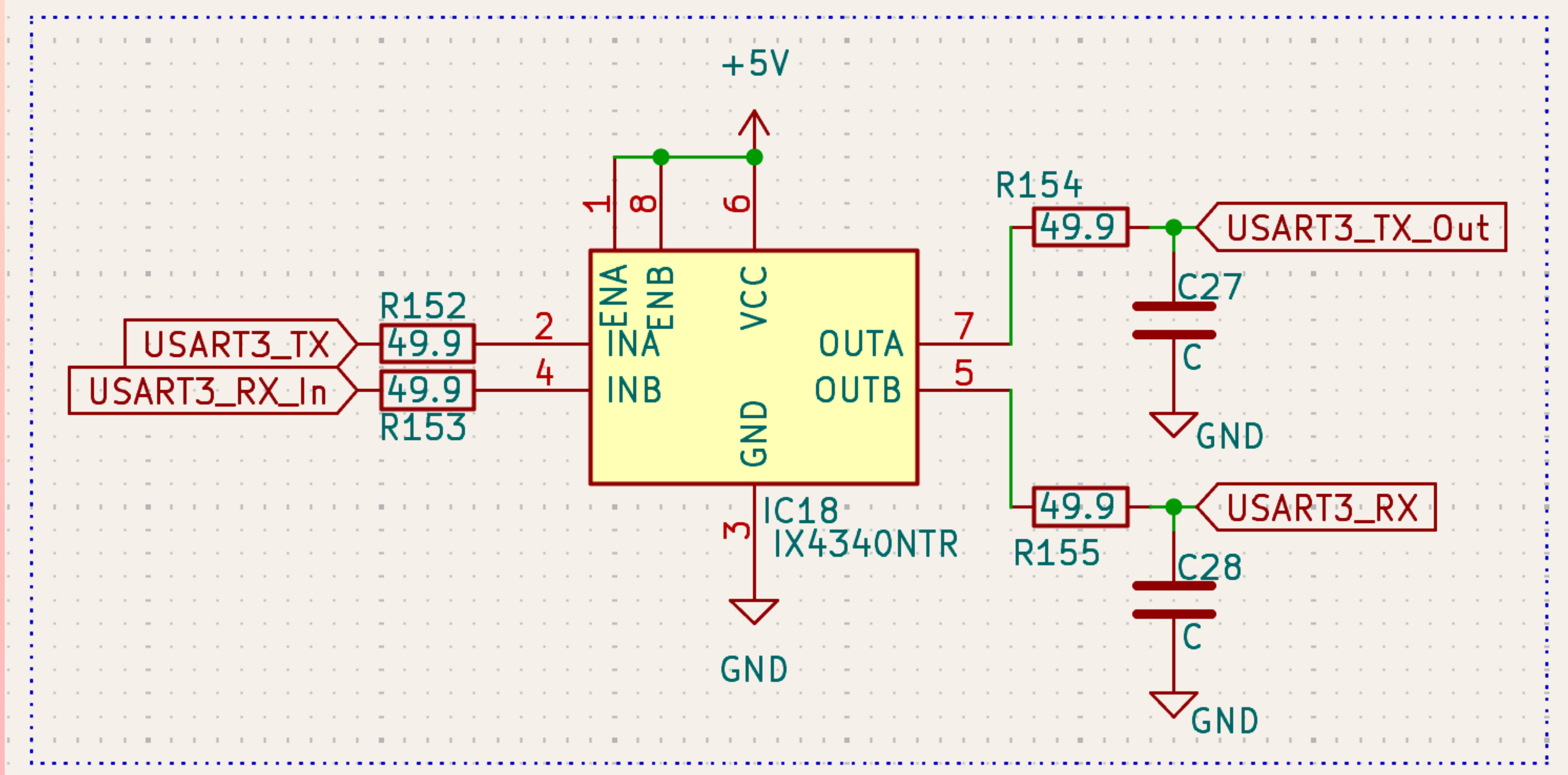
- Select for normally high, low or bypass
- Software fault disable available as well
- Fault handling is physically configurable

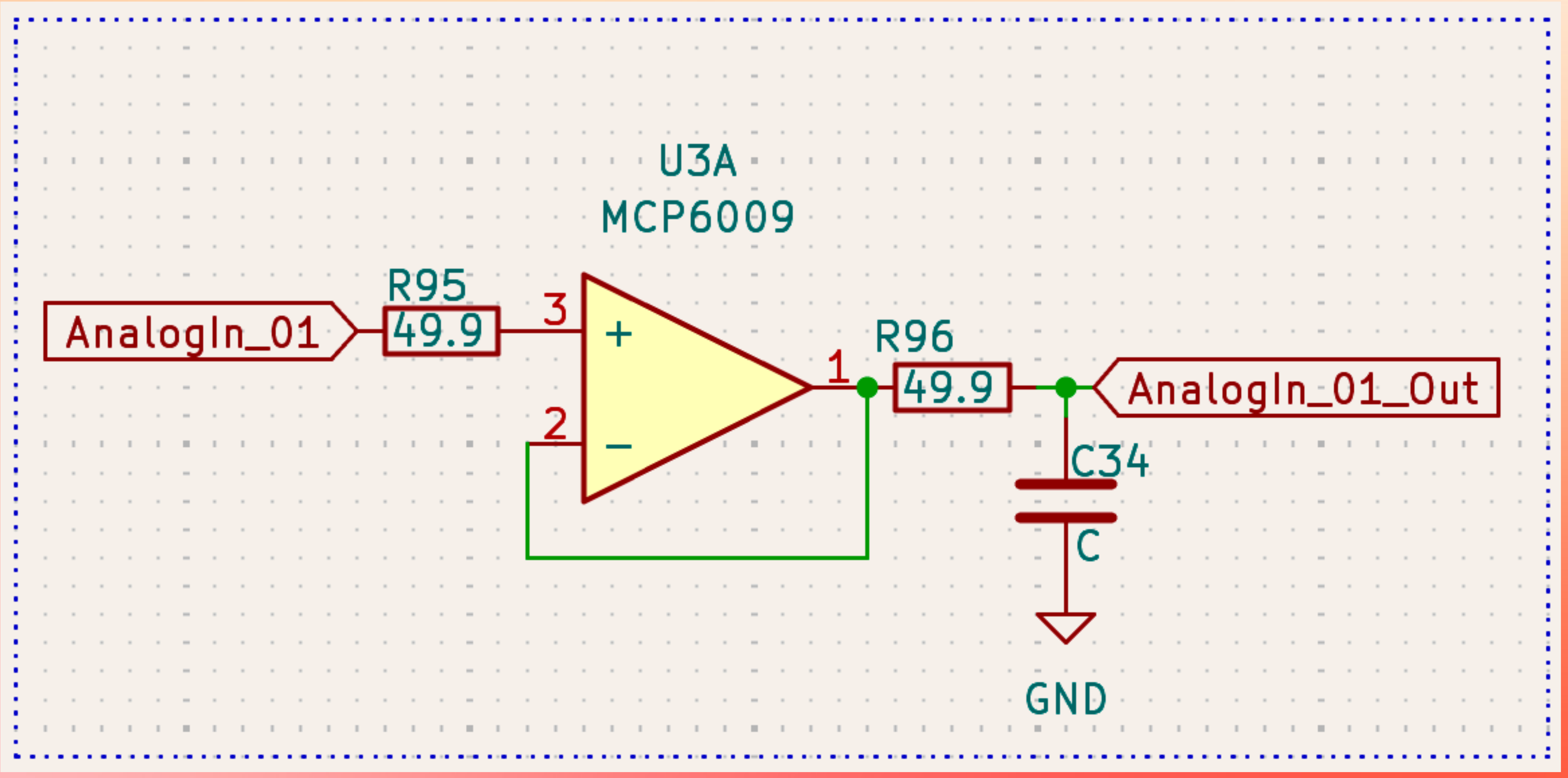


Improved EMI Compatibility



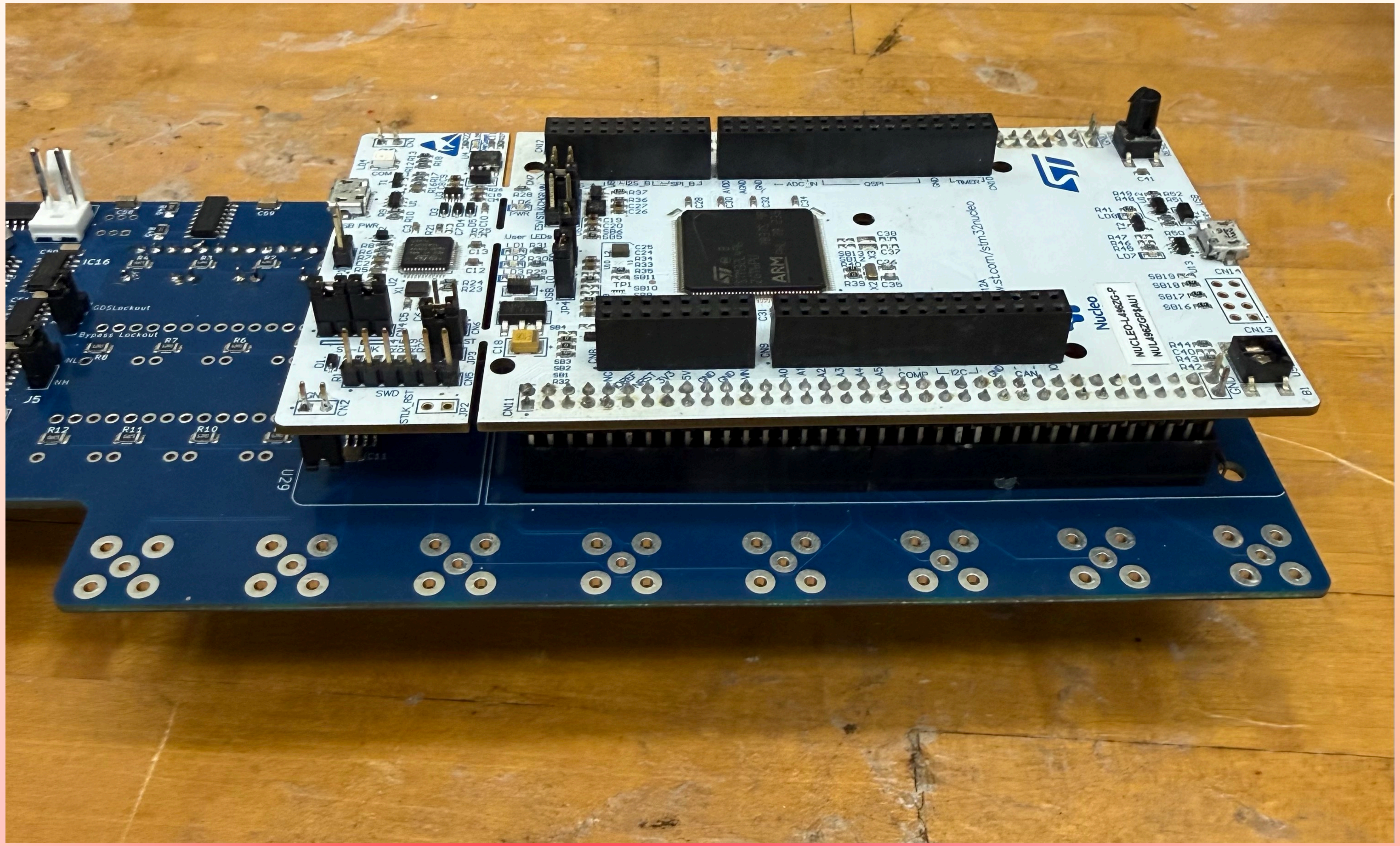
- MCU will be soldered directly to the board
  - No more nucleo board (reduced loop area for all MCU nets)
- Nearly all digital & analog outputs are filtered (exception for programming ports)

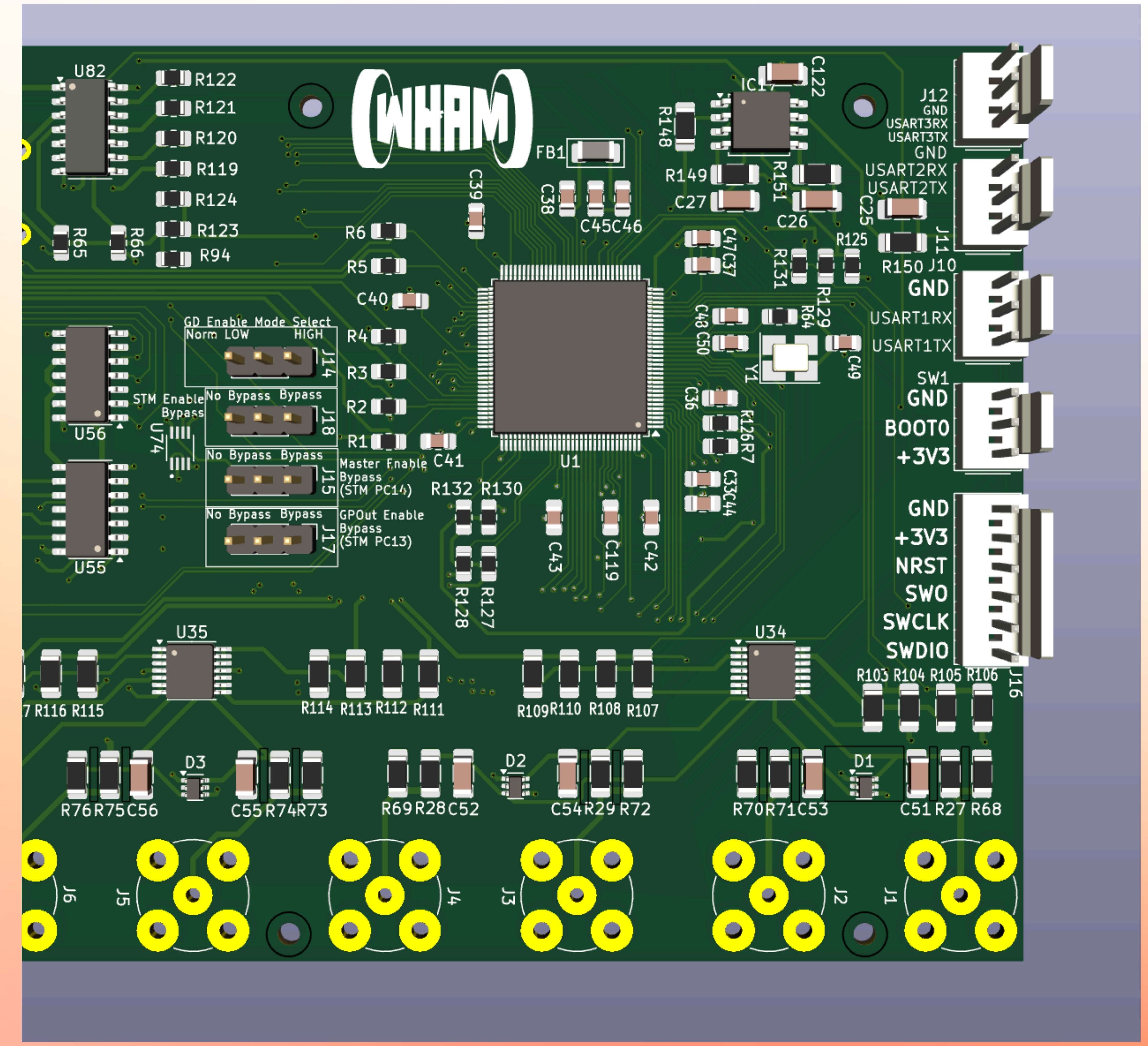




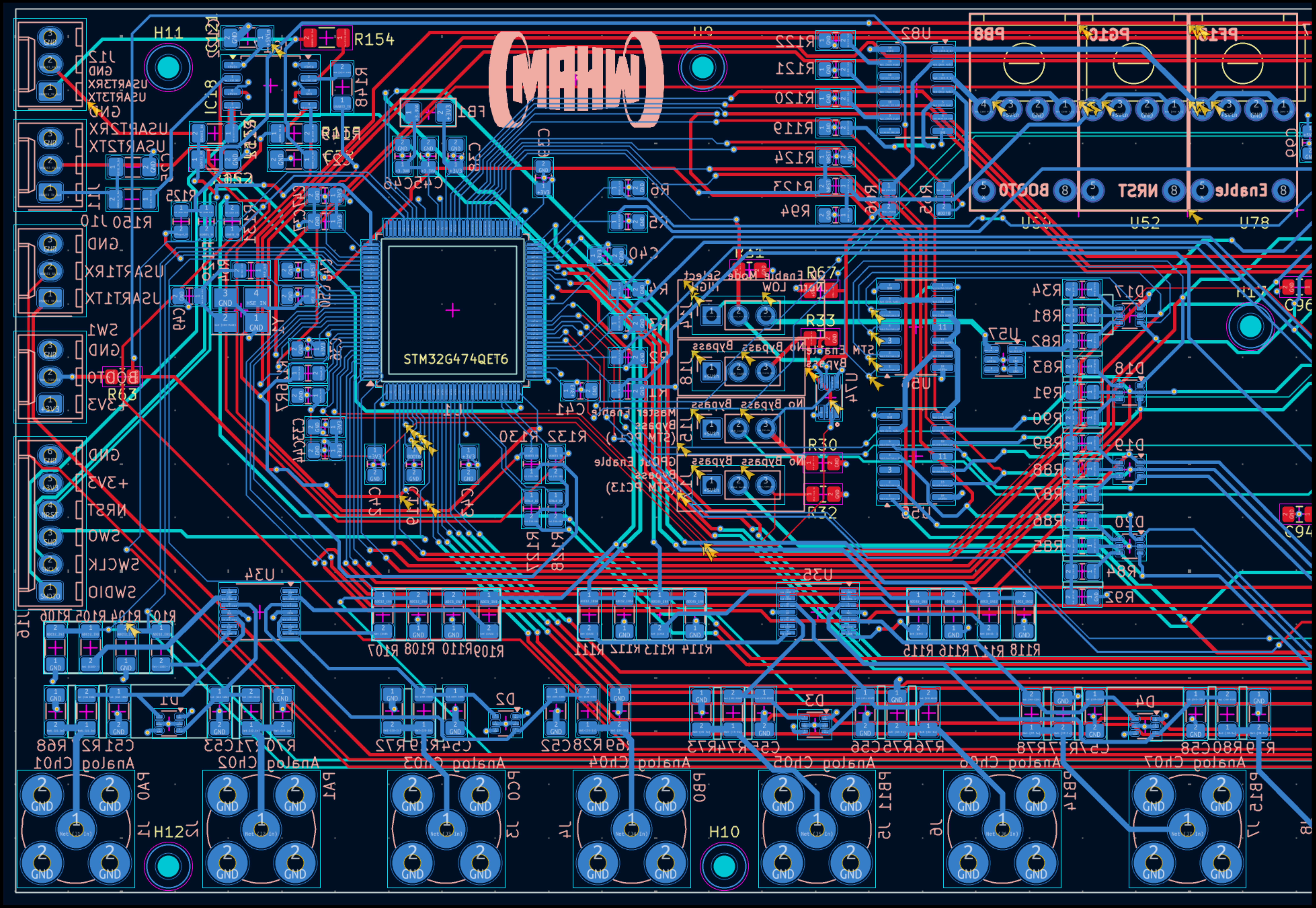
V2 Actual vs V3 Model

Layout






STM32G474QET6
GND
+5V
+3V3
BOOT0
NRST
SWO
SWCLK
SWDIO


H11
R154
FB1
STM32G474QET6
Enable Mode Select
HIGH
LOW
Bypass
No Bypass
Master Enable
GPIO Enable
STM Enable
R67
R33
R30
R32
U82
U57
U34
U35
U53
U52
U78
PB8
PG10
PF10
BOOT0
NRST
Enable
GND
USART1RX
USART1TX
USART2RX
USART2TX
USART3RX
USART3TX
+3V3
SWO
SWCLK
SWDIO
D1
D2
D3
D4
D17
D18
D19
D20
H12
H10
Analog Ch01
Analog Ch02
Analog Ch03
Analog Ch04
Analog Ch05
Analog Ch06
Analog Ch07
PA0
PA1
PC0
PB0
PB11
PB14
PB15
J1
J2
J3
J4
J5
J6
J7

WHAM
+5V
GND
J13
Analog Ch01
Analog Ch02
Analog Ch03
Analog Ch04
Analog Ch05
Analog Ch06
Analog Ch07
Analog Ch08
Phase WN
Phase W
Phase UN
Phase U
Phase VN
Phase V
Enable
NRST
BOOT0
USART1RX
USART1TX
USART2RX
USART2TX
USART3RX
USART3TX
SWDIO
SWCLK
SWO
NRST
+3V3
GND
STM32G474QET6

Other features not mentioned

- Potential to program over ethernet (the hardware is designed in, so this is theoretically possible)
- Dedicated fiber optic ENABLE
- Fiber optic BOOT0 signal (allows for MCU reset with fiber optic signal)
- Dedicated fiber optic channels for serial communication
  - No more designing extra boards to handle this
- Physical hardware is reconfigurable to free up paired PWM fiber outputs for unique purposes
  - Increases flexibility and application for other areas of the lab

Unknowns/Possible Issues

- No built-in ST-LINK
  - Have not attempted programming an MCU using external ST-LINK
  - Design follows ST Application Note 5093 section 6
- HRTIM is designed for 3-phase motor control, power conversion, etc
  - Programming should be simpler than V2 controller, but needs to be put into practice
- PCB design is extremely complicated

HAVE A QUESTION?

I have an answer, let's see if they match!